

Electronics Handbook

2nd Year Electrical Engineering

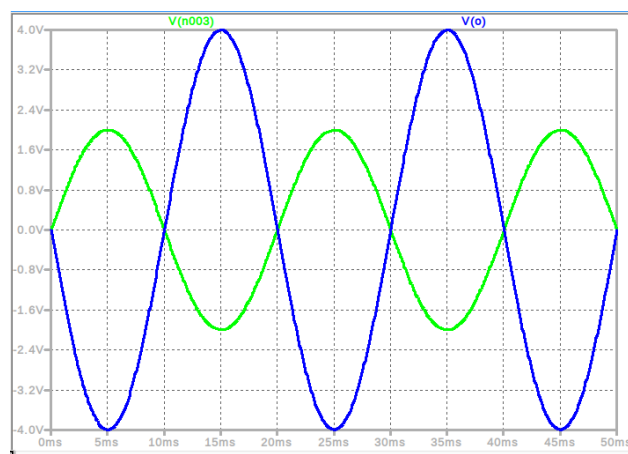
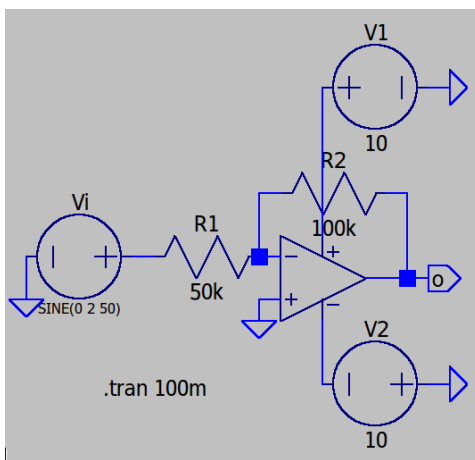
Eng. Abdalrahman Shaban Mohamed

Contents

Chapter 1

Operational Amplifier (Linear Applications)

1.1 Inverting

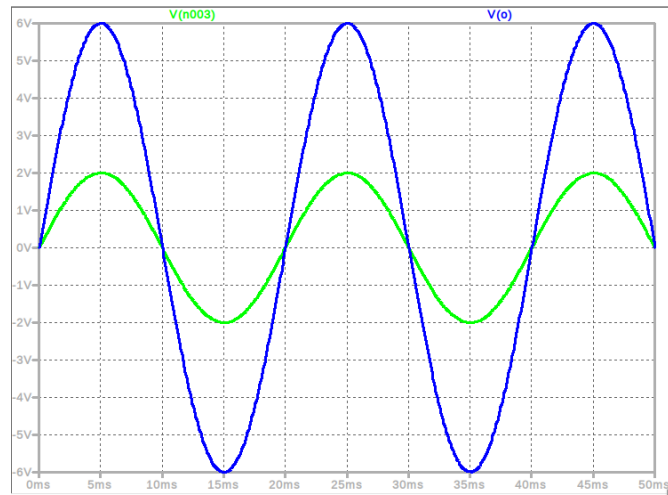
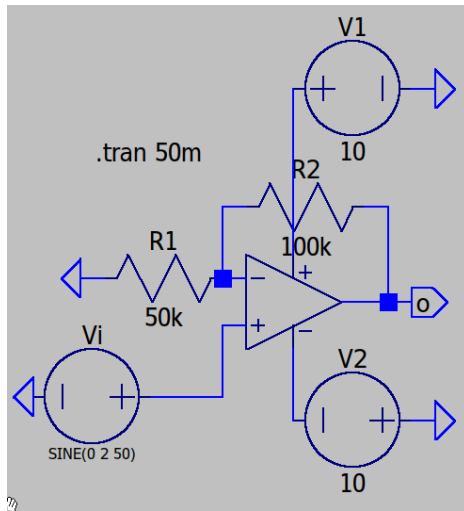


$$\begin{aligned}
 I_{R1} &= I_{R2} \\
 \frac{V_i - 0}{R_1} &= \frac{0 - V_o}{R_2} \\
 R_2 V_i &= -R_1 V_o \\
 \therefore V_o &= -\frac{R_2}{R_1} V_i
 \end{aligned}$$

1.2 Effect of Finite Open-loop Gain

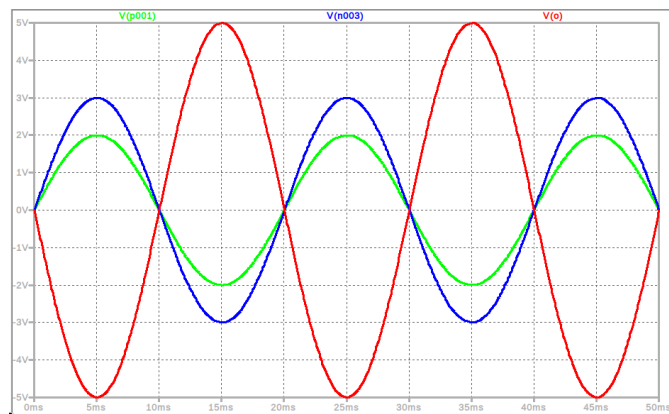
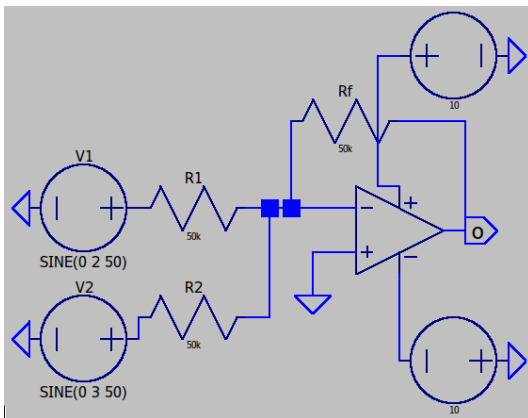
$$\begin{aligned}
 V_o &= A(V^+ - V^-) \\
 V^+ &= 0 \quad \text{and} \quad V^- = -\frac{V_o}{A} \\
 \frac{V_i - (-\frac{V_o}{A})}{R_1} &= \frac{-\frac{V_o}{A} - V_o}{R_2} \\
 R_2 V_i + R_2 \frac{V_o}{A} &= -R_1 \frac{V_o}{A} - R_1 V_o \\
 R_2 V_i &= -V_o \left(\frac{R_2}{A} + \frac{R_1}{A} + R_1 \right) \\
 \frac{R_2}{R_1} V_i &= -V_o \left(\frac{R_2}{AR_1} + \frac{1}{A} + 1 \right) \\
 \therefore G = \frac{V_o}{V_i} &= \frac{-R_2/R_1}{1 + (1 + \frac{R_2}{R_1})/A}
 \end{aligned}$$

1.3 Non-Inverting



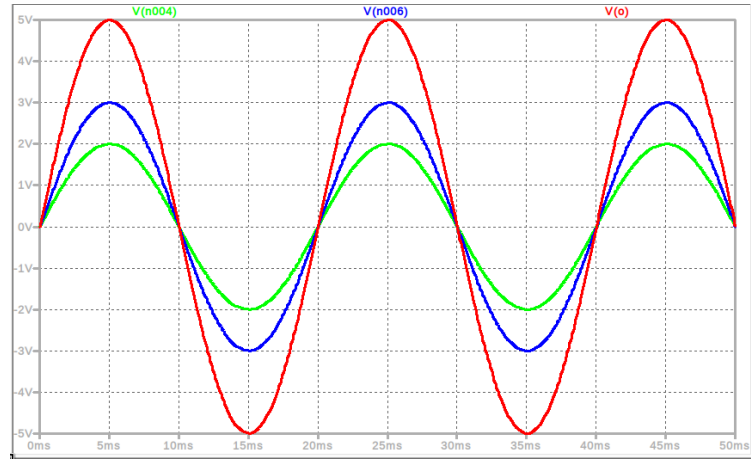
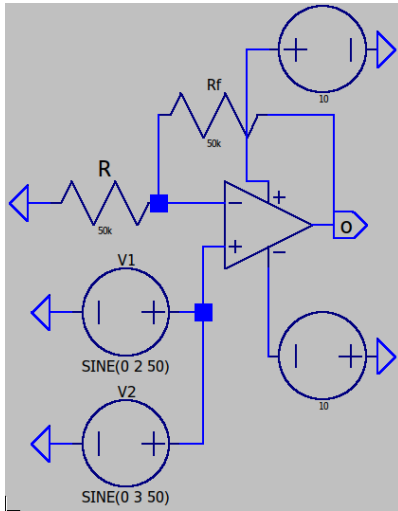
$$\begin{aligned}
 I_{R1} &= I_{R2} \\
 \frac{0 - V_i}{R_1} &= \frac{V_i - V_o}{R_2} \\
 -R_2 V_i &= R_1 V_i - R_1 V_o \\
 -V_i(R_2 + R_1) &= -R_1 V_o \\
 \therefore V_o &= V_i(1 + R_2/R_1)
 \end{aligned}$$

1.4 Inverting Summer



$$\begin{aligned}
 \therefore I_{R1} + I_{R2} &= I_{R_f} \\
 \frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} &= \frac{0 - V_o}{R_f} \\
 \therefore V_o &= -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2\right)
 \end{aligned}$$

1.5 NonInverting Summer



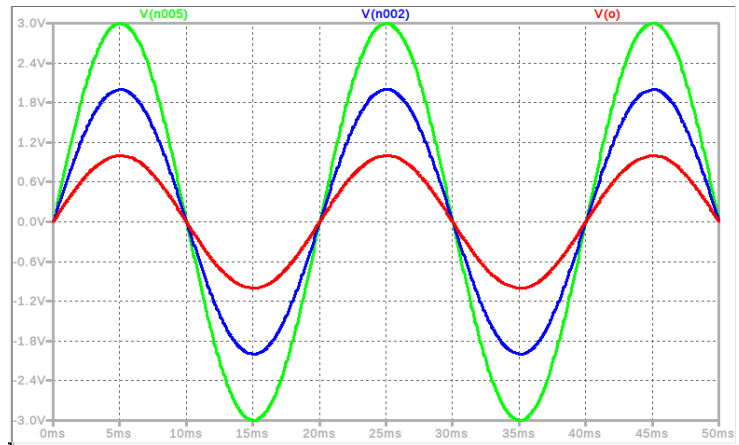
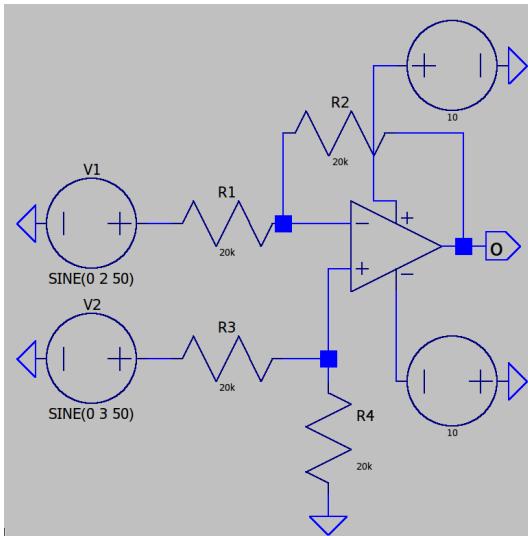
$$\therefore V^- = V^+$$

$$\frac{0 - (V_1 + V_2)}{R} = \frac{(V_1 + V_2) - V_o}{R_f}$$

$$-R_f(V_1 + V_2) = R(V_1 + V_2) - RV_o$$

$$\therefore V_o = (1 + R_f/R)(V_1 + V_2)$$

1.6 Difference



Using Superposition:

kill V_2 :

$$V_{o1} = -\frac{R_2}{R_1}V_1$$

kill V_1 :

$$\begin{aligned} 0 - \frac{R_4}{R_3+R_4}V_2 &= \frac{\frac{R_4}{R_3+R_4}V_2 - V_{o2}}{\frac{R_1}{R_2}} \\ \frac{-R_4R_2}{R_3+R_4}V_2 &= \frac{R_4R_1}{R_3+R_4}V_2 - R_1V_{o2} \\ V_{o2} &= \frac{R_2}{R_1} \frac{R_4}{R_3+R_4}V_2 + \frac{R_4}{R_3+R_4}V_2 \end{aligned}$$

$$V_o = V_{o1} + V_{o2}$$

$$V_o = -\frac{R_2}{R_1}V_1 + \frac{R_2}{R_1} \frac{R_4}{R_3+R_4}V_2 + \frac{R_4}{R_3+R_4}V_2$$

$$V_o = -\frac{R_2}{R_1}V_1 + \frac{R_4}{R_3+R_4} \left(\frac{R_2}{R_1} + 1 \right) V_2$$

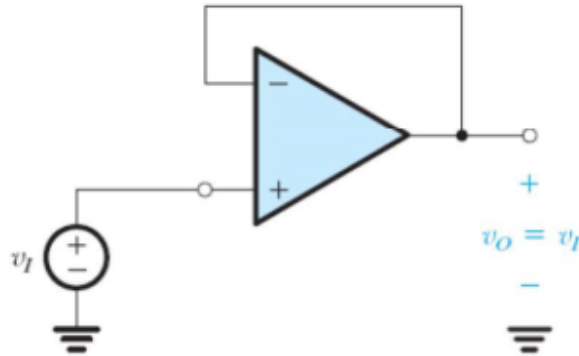
When $R_1 = R_3$, $R_2 = R_4$

$$V_o = -\frac{R_2}{R_1}V_1 + \frac{R_2}{R_1+R_2} \left(\frac{R_2}{R_1} + 1 \right) V_2$$

$$V_o = -\frac{R_2}{R_1}V_1 + \frac{R_2}{R_1+R_2} \left(\frac{R_1+R_2}{R_1} \right) V_2$$

$$\therefore V_o = \frac{R_2}{R_1}(V_2 - V_1)$$

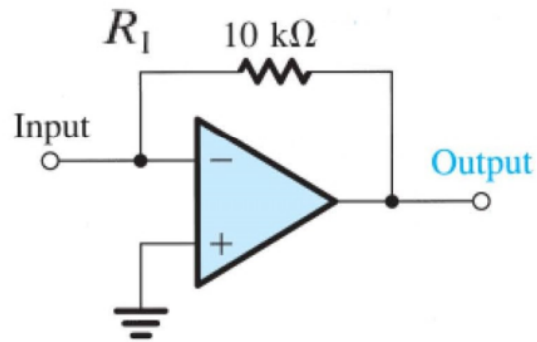
1.7 Buffer



$$V_o = V_i \quad (1.1)$$

The main function of an op-amp buffer is to provide a high input impedance and a low output impedance, which helps to reduce the loading effect on the circuit it is connected to.

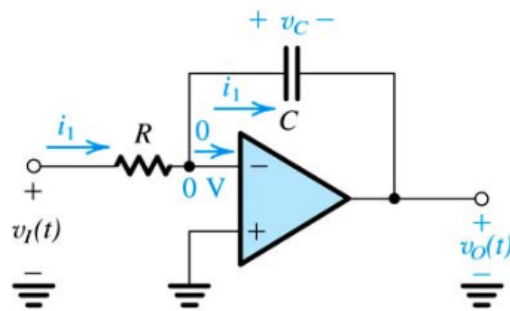
1.8 Trans-Impedance



$$\begin{aligned} I_i &= I_R \\ I_i &= \frac{0 - V_o}{R_f} \\ \therefore V_o &= -R_f I_i \end{aligned}$$

1.9 Integrator

1.9.1 Inverting Integrator



$$\begin{aligned}\frac{V_i - 0}{R} &= I_c \\ \therefore V_o &= -V_c \\ \frac{V_i - 0}{R} &= -C \frac{dV_o}{dt} \\ \therefore V_o &= \frac{-1}{RC} \int V_i dt\end{aligned}$$

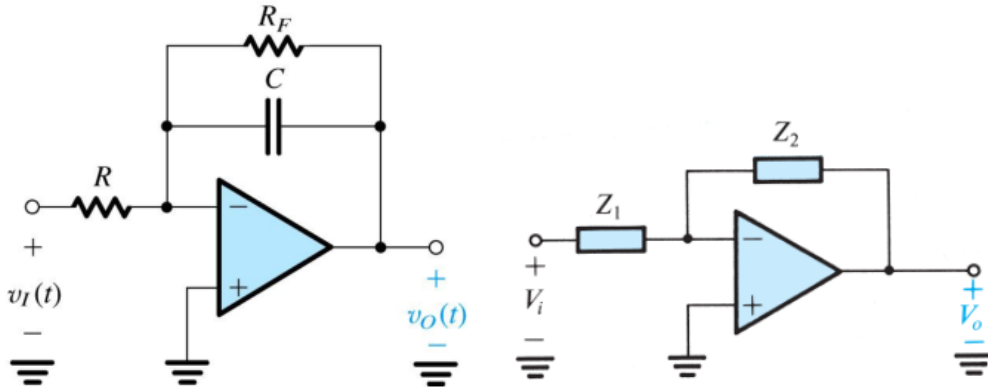
In s domain:

$$\begin{aligned}V_o &= \frac{-1}{sRC} V_i \\ \frac{V_o(j\omega)}{V_i(j\omega)} &= \frac{-1}{(j\omega)RC} \\ \frac{V_o(j\omega)}{V_i(j\omega)} &= \frac{-(-j)}{(\omega)RC}\end{aligned}$$

$$\text{As } j = \cos\left(\frac{\pi}{2}\right) + j\sin\left(\frac{\pi}{2}\right) = e^{j\frac{\pi}{2}}$$

$$\begin{aligned}\frac{V_o(\omega)}{V_i(\omega)} &= \frac{e^{j\frac{\pi}{2}}}{\omega RC} \\ \therefore \left| \frac{V_o}{V_i} \right| &= \left| \frac{1}{\omega RC} \right| \\ \phi &= +90\end{aligned}$$

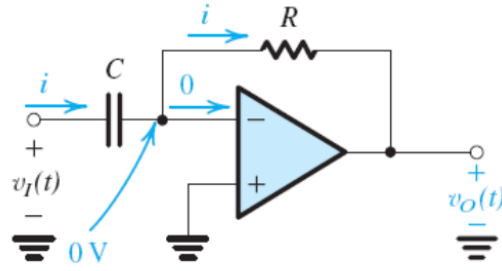
1.9.2 Miller Integrator



The Miller integrator with a large resistance R_F connected in parallel with C in order to provide negative feedback and hence finite gain at dc ($A = \frac{R_f}{R}$).

$$\begin{aligned}V_o &= -\frac{Z_2}{Z_1} \\ Z_2 &= \frac{R_f \cdot \frac{1}{sC}}{R_f + \frac{1}{sC}} \\ Z_2 &= \frac{R_f}{sR_fC + 1} \\ \therefore V_o &= \frac{-R_f/R}{sR_fC + 1} \cdot V_i\end{aligned}$$

1.10 Differentiator

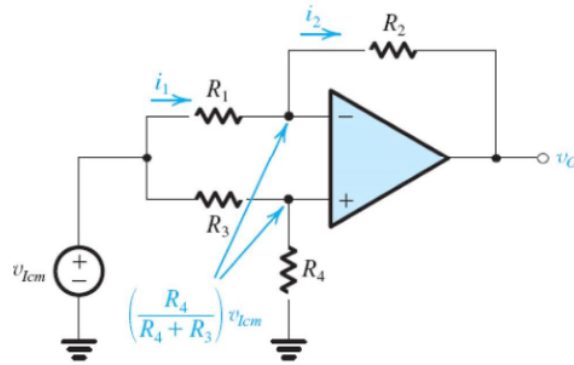


$$I_C = I_R$$

$$C \frac{dV_i}{dt} = \frac{0 - V_o}{R}$$

$$\therefore V_o = -RC \frac{dV_i}{dt}$$

1.11 Common Mode Gain



$$V^- = \frac{R_4}{R_4 + R_3} V_{Icm}$$

$$\frac{V_{Icm} - V^-}{R_1} = \frac{V_{Icm} - V_o}{R_2}$$

$$R_2 V_{Icm} - \frac{R_4 R_2}{R_4 + R_3} V_{Icm} = \frac{R_4 R_1}{R_4 + R_3} V_{Icm} - R_1 V_o$$

$$V_o = \frac{R_2}{R_1} \frac{R_4}{R_4 + R_3} V_{Icm} + \frac{R_4}{R_4 + R_3} V_{Icm} - \frac{R_2}{R_1} V_{Icm}$$

$$\frac{V_o}{V_{Icm}} = \frac{R_4}{R_4 + R_3} \left(\frac{R_2}{R_1} + 1 - \frac{R_4 + R_3}{R_4} \cdot \frac{R_2}{R_1} \right)$$

$$\frac{V_o}{V_{Icm}} = \frac{R_4}{R_4 + R_3} \left(\frac{R_2}{R_1} \left(1 - \frac{R_4 + R_3}{R_4} \right) + 1 \right)$$

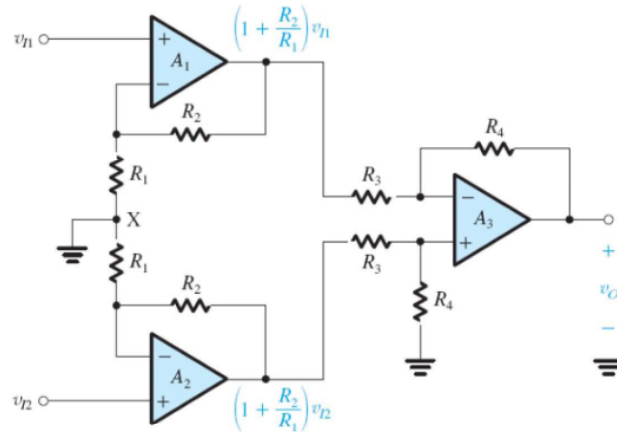
$$\frac{V_o}{V_{Icm}} = \frac{R_4}{R_4 + R_3} \left(\frac{R_2}{R_1} \left(1 - 1 - \frac{R_3}{R_4} \right) + 1 \right)$$

$$\therefore A_{cm} = \frac{V_o}{V_{Icm}} = \frac{R_4}{R_4 + R_3} \left(1 - \frac{R_2 R_3}{R_1 R_4} \right)$$

When $R_3 = R_1$ and $R_4 = R_2$
 $A_{cm} = 0$

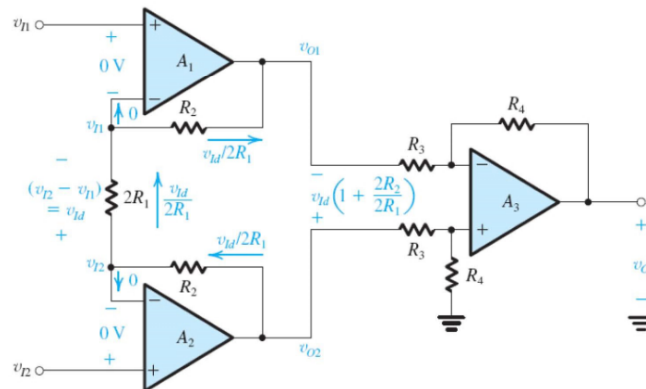
1.12 Instrumentation

To get high input impedance and zero common mode gain (less noise) we can use instrumentation Amplifier.



What're the major disadvantages of this configuration?

- 1- Saturation due to the common mode voltage.
- 2- Common mode voltage will be amplified unless we make perfect matching between the resistors which is very hard in practical.
- 3- We have to change at least 2 resistors to change the gain.



$$A_{cm} = 0$$

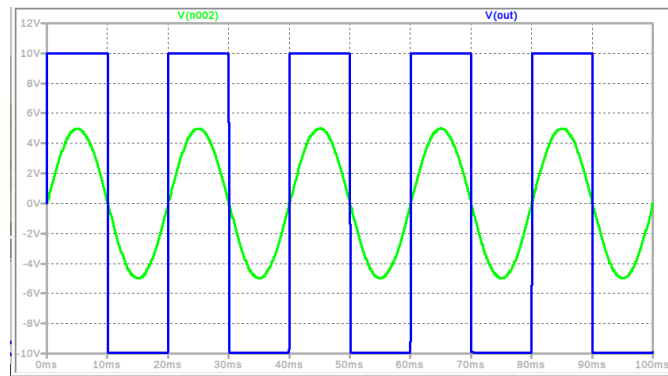
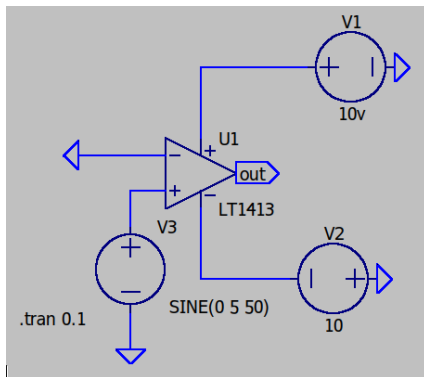
$$A_d = \frac{R_4}{R_3} \left(1 + \frac{2R_2}{2R_1} \right)$$

Chapter 2

Operational Amplifier (Non-Linear Applications)

2.1 Comparator

2.1.1 Zero-level detector

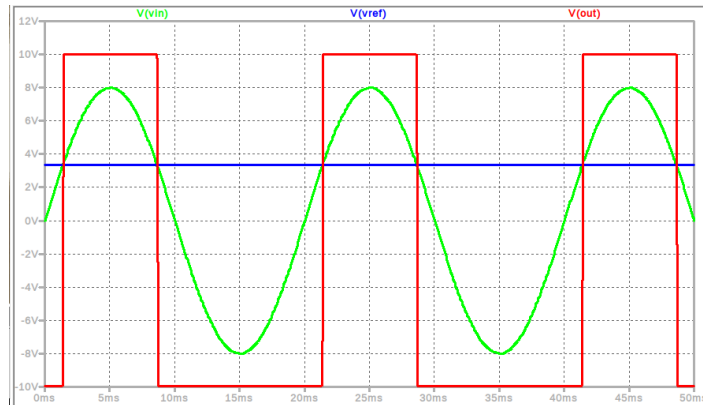
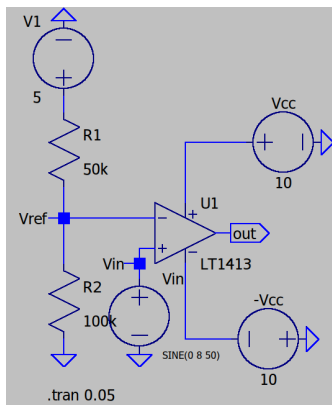


$$V_o = A(V^+ - V^-)$$

$$\because V^- = 0$$

$$V_o = V_{cc}$$

2.1.2 NonZero-level detector



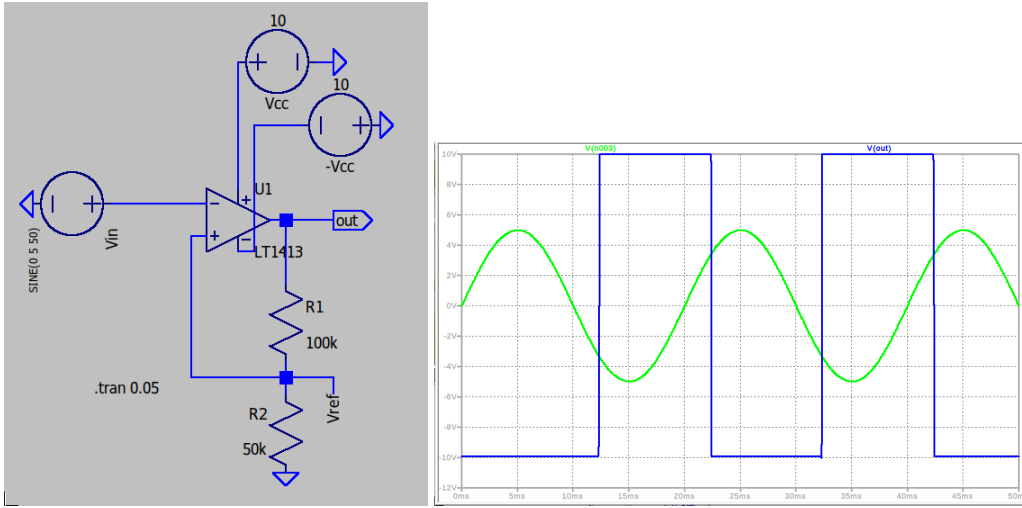
$$\because V_o = A(V^+ - V^-)$$

$$\because V^- = V_{ref} = V_1 \cdot \frac{R_2}{R_2 + R_1}$$

$$\therefore V_o = +V_{cc} \quad \text{when} \quad V^+ > V_{ref}$$

$$\therefore V_o = -V_{cc} \quad \text{when} \quad V^+ \leq V_{ref}$$

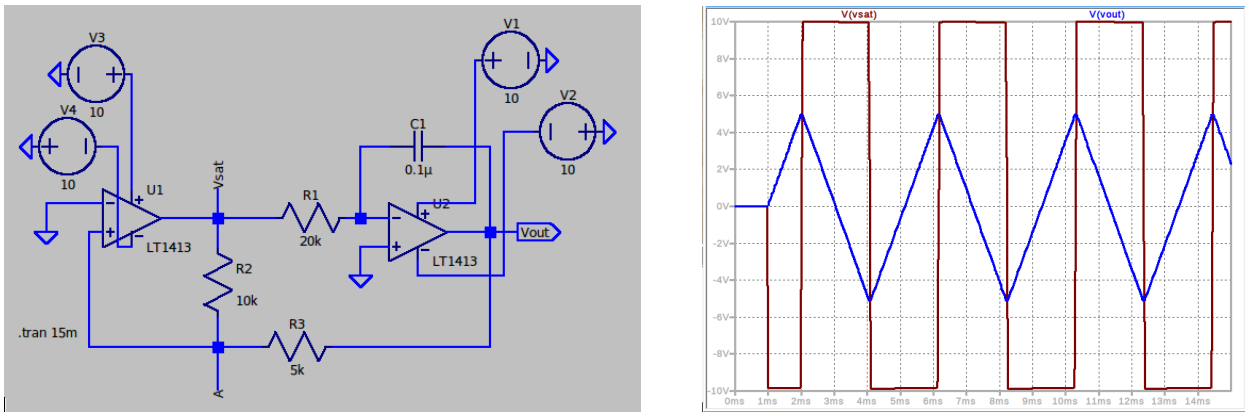
2.1.3 Schmitt Trigger



Using Schmitt Trigger reduces noise effects with hysteresis.

$$\begin{aligned} \because V_o &= A(V^+ - V^-) \\ &= V_o = A(V_{ref} - V_{in}) \\ \text{As } V_{ref} &= V_{cc} \cdot \frac{R_2}{R_2 + R_1} \end{aligned}$$

2.2 Practical Triangular-Wave Oscillator



Let V_{UTP} be the Upper Trigger Point (UTP) (V_{out})

And V_{LTP} be the Lower Trigger Point (LTP)

KCL at pint A:

$$\begin{aligned} \frac{V_{out} - V_A}{R_3} &= \frac{V_A - V_{sat}}{R_2} \\ \text{As } V_{out} &= V_{UTP} = |V_{LTP}| \\ R_2 V_o - R_2 V_A &= R_3 V_A - R_3 V_{sat} \\ V_o &= \frac{R_3}{R_2} V_A - \frac{R_3}{R_2} V_{sat} + V_A \\ &= \frac{R_3}{R_2} (V_A - V_{sat}) + V_A \\ \text{At } V_A &= 0 \\ V_o &= -V_{sat} \frac{R_3}{R_2} \end{aligned}$$

So we can find out that:

$$V_{UTP} = \frac{R_3}{R_2} V_{sat}$$

$$V_{LTP} = -\frac{R_3}{R_2} V_{sat}$$

4 of the capacitor charging time is the Periodic time.

$$V_{UTP} = \frac{1}{R_1 C} \int dV_{sat} dt$$

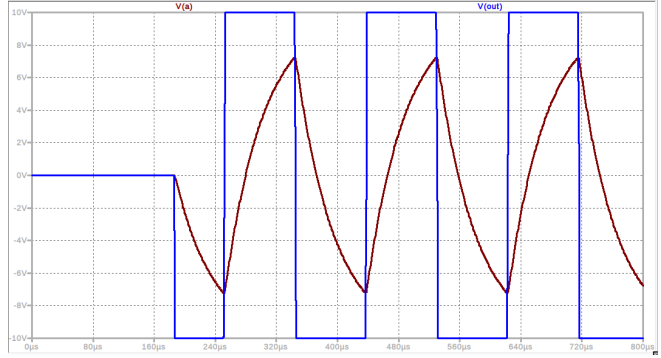
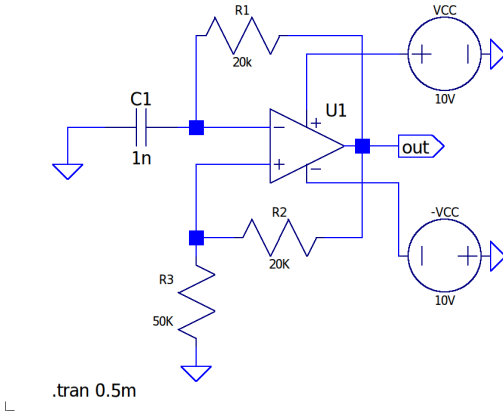
$$\frac{R_3}{R_2} V_{sat} = \frac{1}{R_1 C} V_{sat} t$$

$$t = R_1 C \frac{R_3}{R_2}$$

$$T = 4t = 4 \frac{R_3}{R_2} R_1 C$$

$$\therefore F = \frac{1}{T} = \frac{1}{4R_1 C} \cdot \frac{R_2}{R_3}$$

2.3 Square-Wave Oscillator (Astable Multivibrator)



When $V_c = 0$ and Capacitor begins charging

$$\therefore V_o = A \cdot (V_f - V_c)$$

$$\therefore V_f > V_c$$

$$\therefore V_o = V_{sat}$$

When $V_c \geq V_f$

$$V_o = -V_{sat}$$

$$\therefore V_f = \beta V_{sat} \quad \text{As } \beta = \frac{R_3}{R_3 + R_2}$$

Using KCL

$$\frac{V_{sat} - V_c}{R_1} = C \frac{dV_c}{dt}$$

$$\frac{V_{sat} - V_c}{R_1 C} = \frac{dV_c}{dt}$$

$$\int_0^t \frac{dt}{R_1 C} = \int_{-V_f}^{V_f} \frac{dV_c}{V_{sat} - V_c}$$

$$\frac{t}{R_1 C} = \ln(V_{sat} + V_f) - \ln(V_{sat} - V_f)$$

$$\frac{t}{R_1 C} = \ln\left(\frac{V_{\text{sat}} + V_f}{V_{\text{sat}} - V_f}\right)$$

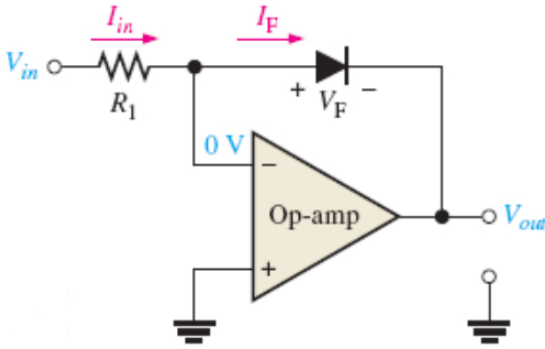
$$t = R_1 C \cdot \ln\left(\frac{1 + \beta}{1 - \beta}\right)$$

$$T = 2t = 2R_1 C \cdot \ln\left(\frac{1 + \beta}{1 - \beta}\right)$$

$$\therefore F = \frac{1}{2R_1 C \cdot \ln\left(\frac{1 + \beta}{1 - \beta}\right)}$$

2.4 Log and Antilog Amplifier

2.4.1 Basic log Amplifier



$$I_D = I_F = I_R e^{\frac{V_F}{V_T}}$$

I_R is const

$$V_T = \frac{q}{kt} \approx 26mV$$

$$V_o = -V_F$$

Using KCL:

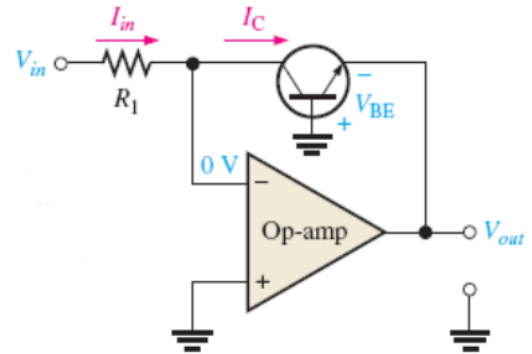
$$I_{\text{in}} = I_F$$

$$\frac{V_{\text{in}}}{R_1} = I_R e^{\frac{-V_o}{V_T}}$$

$$\frac{-V_o}{V_T} = \ln\left(\frac{V_{\text{in}}}{R_1 I_R}\right)$$

$$\therefore V_o = -V_T \ln\left(\frac{V_{\text{in}}}{R_1 I_R}\right)$$

2.4.2 Log Amplifier with BJT



$$I_E = I_C + I_B$$

$$I_C \gg I_B$$

$$I_E \approx I_C$$

$$I_C = I_{\text{EBO}} e^{\frac{V_{\text{BE}}}{V_T}}$$

$$I_{\text{in}} = I_C$$

$$\frac{V_i}{R_1} = I_{\text{EBO}} e^{\frac{-V_o}{V_T}}$$

$$\therefore V_o = -V_T \ln\left(\frac{V_{\text{in}}}{R_1 I_{\text{EBO}}}\right)$$

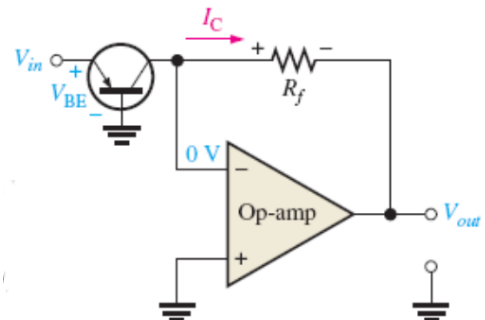
2.4.3 Basic Antilog

$$V_o = -R_f I_C$$

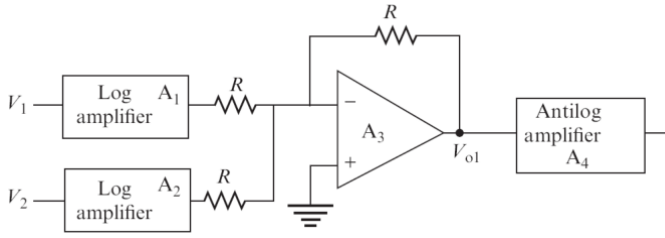
$$I_C = I_{\text{EBO}} e^{\frac{V_i}{V_T}}$$

$$\therefore V_o = -R_f I_{\text{EBO}} e^{\frac{V_i}{V_T}}$$

$$V_o = -R_f I_{\text{EBO}} \text{antilog}\left(\frac{V_i}{26mV}\right)$$

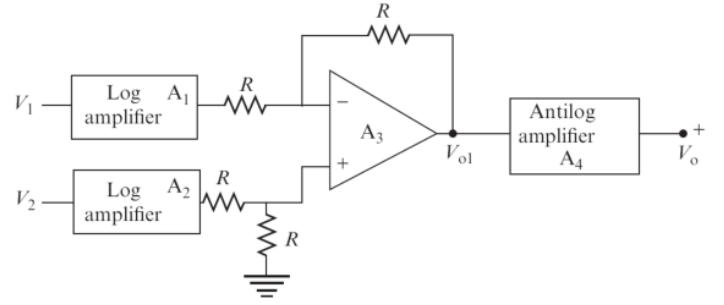


2.5 Multiplier



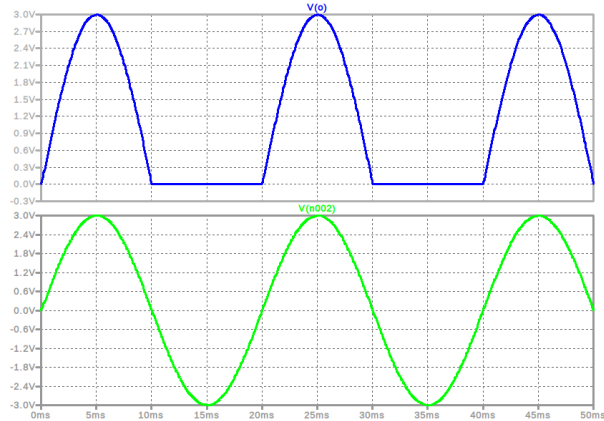
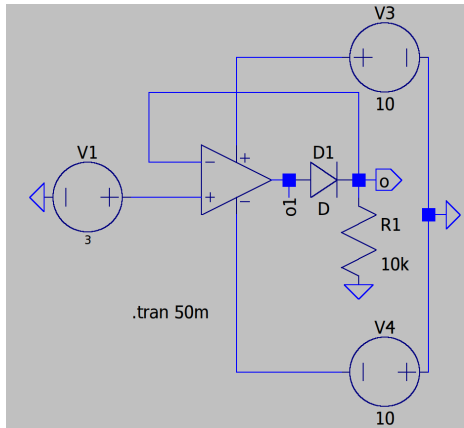
$$\begin{aligned}
 V_{o1} &= V_T \left(\ln\left(\frac{V_1}{RI_{EBO}}\right) + \ln\left(\frac{V_2}{RI_{EBO}}\right) \right) \\
 &= V_T \left(\ln\left(\frac{V_1 V_2}{R^2 I_{EBO}^2}\right) \right) \\
 V_o &= -RI_{EBO} e^{\ln\left(\frac{V_1 V_2}{R^2 I_{EBO}^2}\right)} \\
 &= -RI_{EBO} \frac{V_1 V_2}{R^2 I_{EBO}^2} \\
 \therefore V_o &= -\frac{V_1 V_2}{RI_{EBO}}
 \end{aligned}$$

2.6 Divider



$$\begin{aligned}
 V_{o1} &= -V_T \ln\left(\frac{V_2}{RI_{EBO}}\right) + V_T \ln\left(\frac{V_1}{RI_{EBO}}\right) \\
 &= V_T \left(\ln\left(\frac{V_1}{V_2}\right) \right) \\
 V_o &= -RI_{EBO} e^{\ln\left(\frac{V_1}{V_2}\right)} \\
 \therefore V_o &= -RI_{EBO} \frac{V_1}{V_2}
 \end{aligned}$$

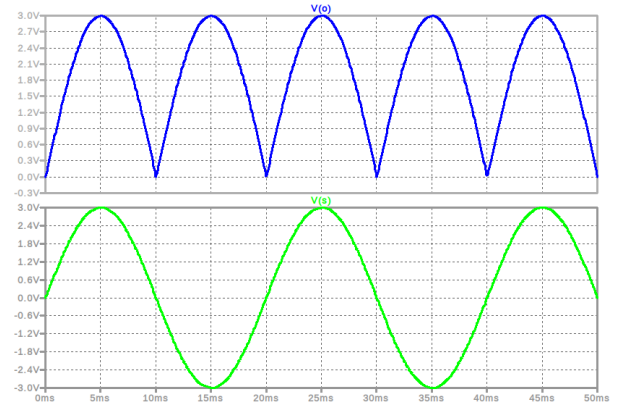
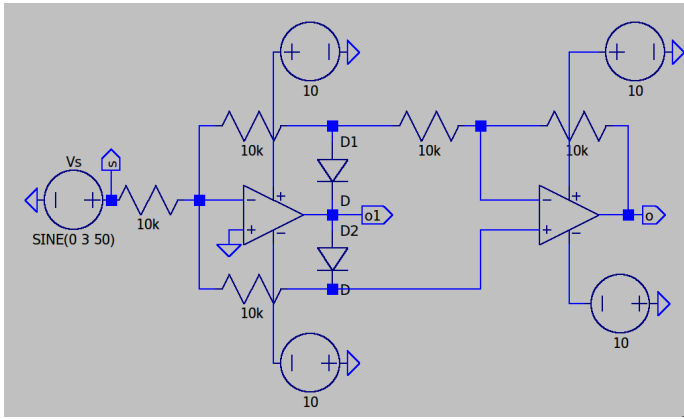
2.7 Positive half-wave rectifier



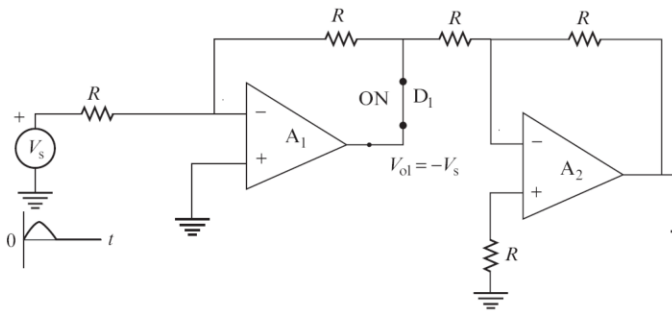
$$V_{o1} \approx V_1 + 0.623$$

$$V_o = V_1 \text{ Rectified}$$

2.8 Full-wave Rectifier

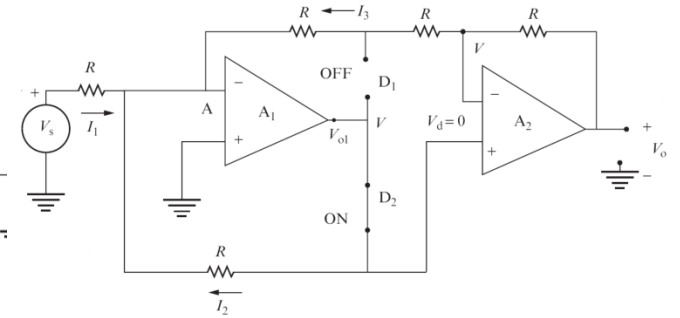


During positive half cycle



$$\begin{aligned}\frac{V_s - o}{R} &= \frac{0 - V_{o1}}{R} \\ V_{o1} &= -V_s \\ \frac{V_{o1} - o}{R} &= \frac{0 - V_o}{R} \\ V_o &= -V_{o1} \\ \therefore V_o &= V_s\end{aligned}$$

During negative half cycle

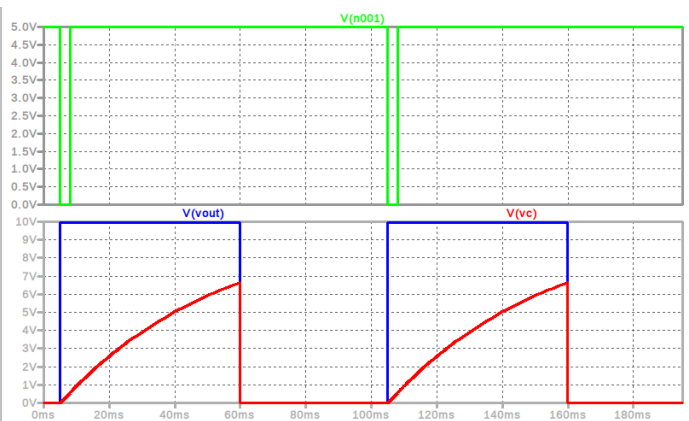
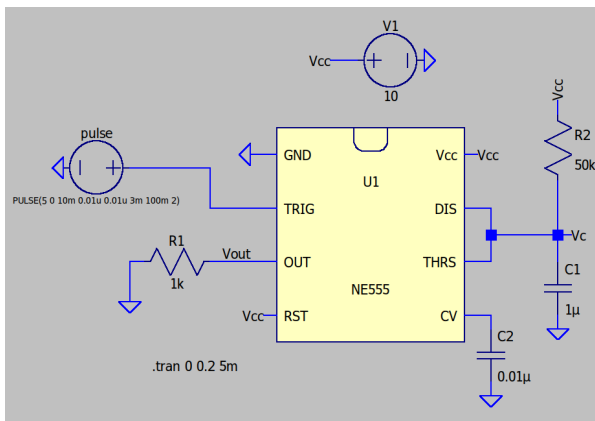
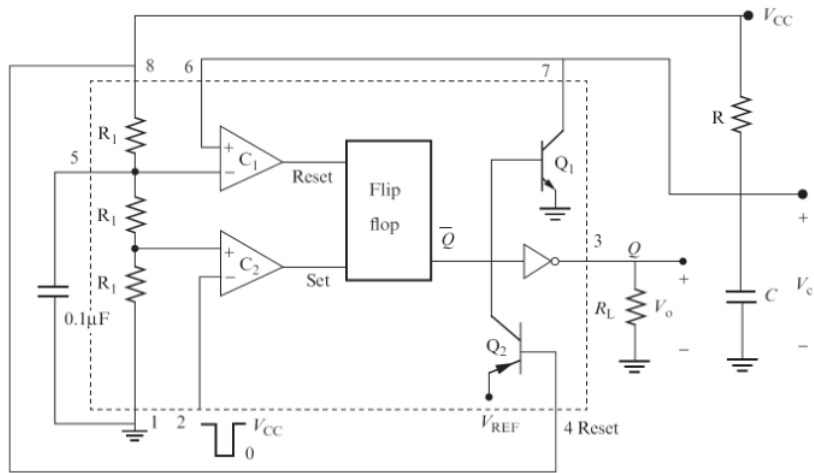


$$\begin{aligned}I_1 + I_2 + I_3 &= 0 \\ \frac{V_s}{R} + \frac{V}{R} + \frac{V}{2R} &= 0 \\ V &= -\frac{2}{3}V_s \\ \frac{0 - V}{2R} &= \frac{V - V_o}{R} \\ V_o &= \frac{3}{2}V \\ V_o &= \frac{3}{2} \left(-\frac{2}{3}V_s \right) \\ \therefore V_o &= -V_s\end{aligned}$$

Chapter 3

Timer 555

3.1 Monostable Multivibrator



$$V_c(t) = V_F - (V_F - V_I)e^{\frac{-t}{\tau}}$$

$$V_c(t) = V_{cc} - (V_{cc} - o)e^{\frac{-t}{RC}}$$

$$\text{At } t = T, V_c(t) = \frac{2}{3}V_{cc}$$

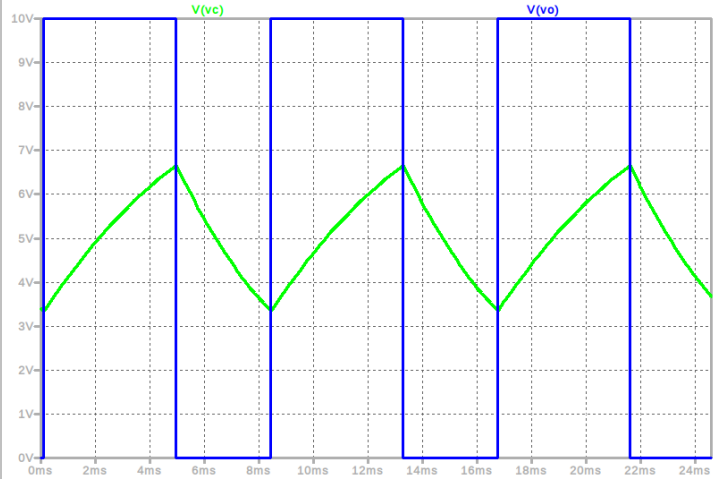
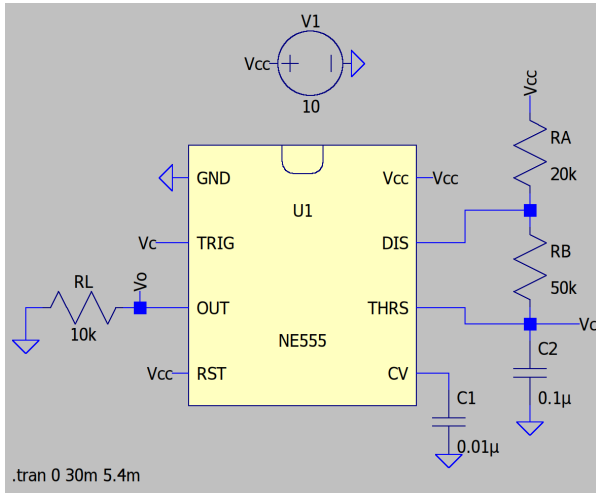
$$\frac{2}{3}V_{cc} = V_{cc}\left(1 - e^{\frac{-T}{RC}}\right)$$

$$e^{\frac{-T}{RC}} = \frac{1}{3}$$

$$\frac{-T}{RC} \approx -1.1$$

$$\therefore T = 1.1RC$$

3.2 Astable Multivibrator



$$V_c(t) = V_F - (V_F - V_I)e^{-\frac{t}{\tau}}$$

As V_c is applied to Trigger and Threshold:

$$V_F = V_{cc} \text{ and } V_i = \frac{1}{3}V_{cc}$$

$$V_c(t) = V_{cc} - (V_{cc} - \frac{1}{3}V_{cc})e^{-\frac{t}{\tau_c}}$$

$$V_c(t) = \frac{2}{3}V_{cc}$$

$t = T_1 \rightarrow$ Time of charging

$$\frac{2}{3}V_{cc} = V_{cc} - \left(V_{cc} - \frac{1}{3}V_{cc}\right)e^{-\frac{T_1}{\tau_c}}$$

$$\frac{2}{3} = 1 - \frac{2}{3}e^{-\frac{T_1}{\tau_c}}$$

$$\frac{1}{2} = e^{-\frac{T_1}{\tau_c}}$$

$$-0.693 = -\frac{T_1}{\tau_c}$$

$$\tau_c = (R_A + R_B)C$$

$$\therefore T_1 = 0.693\tau_c = 0.693(R_A + R_B)C$$

Calculation of T_2 :

$$V_c(t) = V_F - (V_F - V_I)e^{-\frac{t}{\tau_d}}$$

$$V_F = 0 \text{ and } V_i = \frac{2}{3}V_{cc}$$

$$V_c(t) = \frac{1}{3}V_{cc}$$

$t = T_2 \rightarrow$ Time of discharging

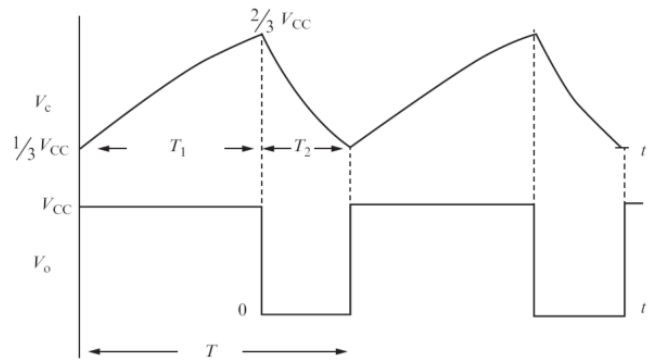
$$\frac{1}{3}V_{cc} = \frac{2}{3}V_{cc}e^{-\frac{t}{\tau_d}}$$

$$\frac{1}{2} = e^{-\frac{t}{\tau_d}}$$

$$0.693 = \frac{t}{\tau_d}$$

$$\tau_d = R_B C$$

$$\therefore T_2 = 0.693R_B C$$



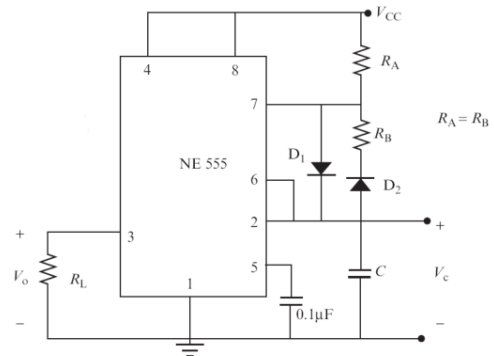
$$T = T_1 + T_2$$

$$= 0.693(R_A + R_B)C + 0.693R_B C$$

$$\therefore T = 0.693(R_A + 2R_B)C$$

$$f = \frac{1}{T}$$

$$\text{Duty cycle} = D = \frac{T_1}{T}$$



To get 50% Duty cycle

$$T_1 = 0.693R_A C$$

$$T_2 = 0.693R_B C$$

If $R_A = R_B$

$$D = \frac{T_1}{T_1 + T_2} \cdot 100 = 50\%$$